Particle Spectrum of General Six-Derivative Gravity on Flat and de Sitter Backgrounds

Computed with TensorGR.jl[1]

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We present the complete tree-level particle spectrum for the most general parity-even gravitational action with up to six derivatives of the metric in four spacetime dimensions, on both flat Minkowski and maximally symmetric (de Sitter/anti-de Sitter) backgrounds. The flat-space propagator is expressed in terms of spin-projection operators and two form factors $f_2(k^2)$, $f_0(k^2)$ that are quadratic polynomials in the squared momentum. On a maximally symmetric background with cosmological constant Λ, the spectrum is parametrised by an effective Newton constant $\kappa_{\rm eff}$, a massive spin-2 mass m_g^2, and a spin-0 mass m_s^2, all receiving Λ-dependent corrections from the cubic curvature sector. The Bueno–Cano parameters (a, b, c, e) for all six independent cubic curvature invariants are computed numerically and verified analytically. All equations are checked against local copies of the reference papers.

I. INTRODUCTION

Higher-derivative extensions of general relativity have attracted sustained interest as effective field theories of quantum gravity, as candidates for ultraviolet-complete gravitational theories, and as tools for understanding the interplay between unitarity and renormalisability [1–3]. The general parity-even action with up to six derivatives of the metric in four dimensions reads

$$S = \int \mathrm{d}^4x\,\sqrt{g}\,\big[\Lambda_{\rm cc} + \kappa R + \alpha_1 R^2 + \alpha_2 R_{\mu\nu}R^{\mu\nu} + \alpha_3\,\mathcal{G} + \beta_1 R\Box R + \beta_2 R_{\mu\nu}\Box R^{\mu\nu} + \sum_{i=1}^{6}\gamma_i\,\mathcal{I}_i\big], \tag{1}$$

where $\mathcal{G} = R^2_{\mu\nu\rho\sigma} - 4R^2_{\mu\nu} + R^2$ is the Gauss–Bonnet density (topological in 4D on flat backgrounds), and $\mathcal{I}_1, \ldots, \mathcal{I}_6$ are the six independent cubic curvature invariants listed in Table I.

All results presented here were computed with TensorGR.jl, a Julia package for abstract tensor algebra equivalent to `xAct` for *Mathematica*. Every equation has been verified against a local copy of the cited reference paper.

II. FLAT-SPACE PROPAGATOR

On a flat Minkowski background $\bar{g}_{\mu\nu} = \eta_{\mu\nu}$, the cubic curvature terms $\gamma_i\mathcal{I}_i$ do not contribute to the linearised action (they require at least one background curvature factor). The quadratic action is bilinear in the metric perturbation $h_{\mu\nu}$, and the saturated propagator decomposes into spin sectors [4]:

$$\boxed{G_{\mu\nu\alpha\beta}(k) = \frac{P^{(2)}_{\mu\nu\alpha\beta}}{k^2\,f_2(k^2)} - \frac{P^{(0\text{-}s)}_{\mu\nu\alpha\beta}}{2\,k^2\,f_0(k^2)}} \tag{2}$$

where $P^{(2)}$ and $P^{(0\text{-}s)}$ are the Barnes–Rivers spin-2 and spin-0 projection operators [5], and the form factors are

$$f_2(z) = 1 - \frac{\alpha_2}{\kappa}\,z - \frac{\beta_2}{\kappa}\,z^2, \tag{3}$$

$$f_0(z) = 1 + \frac{6\alpha_1 + 2\alpha_2}{\kappa}\,z + \frac{6\beta_1 + 2\beta_2}{\kappa}\,z^2. \tag{4}$$

The spin-1 sector vanishes identically (linearised diffeomorphism invariance). These form factors are derived from the 4D identity $R^2_{\mu\nu} = \frac{1}{2}C^2 + \frac{1}{3}R^2$ (modulo the topological Gauss–Bonnet) by mapping to the canonical Weyl/scalar decomposition [4].

A. Stelle limit ($\beta_i = 0$)

Setting $\beta_1 = \beta_2 = 0$ reduces the form factors to linear polynomials, recovering the fourth-derivative (Stelle) gravity spectrum [1]:

$$m_g^2 = \frac{\kappa}{\alpha_2} \qquad \text{(massive spin-2 pole)}, \tag{5}$$

$$m_s^2 = -\frac{\kappa}{6\alpha_1 + 2\alpha_2} \qquad \text{(spin-0 pole)}. \tag{6}$$

The massive spin-2 state is a ghost when $\alpha_2 > 0$.

B. Six-derivative spectrum

For general (β_1, β_2), the form factors are quadratic in $z = k^2$. Each has two roots, giving up to two massive poles per sector (plus the massless graviton at $k^2 = 0$). The spin-2 masses satisfy

$$\beta_2\,k^4 + \alpha_2\,k^2 - \kappa = 0. \tag{7}$$

When the discriminant $\alpha_2^2 + 4\beta_2\kappa < 0$ (requiring $\beta_2 < 0$), the roots form a complex conjugate pair — the *Lee–Wick* scenario [4].

C. Residue sum rule

The partial-fraction decomposition of the propagator satisfies [4]

$$\frac{1}{z\,f(z)} = \frac{A}{z} + \sum_i \frac{B_i}{z - z_i}, \qquad A + \sum_i B_i = 0, \tag{8}$$

where the sum rule $A+\sum B_i = 0$ follows from the $z \to \infty$ behaviour. This has been verified numerically at 50+ random parameter points.

III. DE SITTER SPECTRUM

On a maximally symmetric background with $\bar{R}_{\mu\nu} = \Lambda\,\bar{g}_{\mu\nu}$, $\bar{R} = 4\Lambda$, $\bar{R}_{\mu\nu\rho\sigma} = \frac{2\Lambda}{3}(\bar{g}_{\mu\rho}\bar{g}_{\nu\sigma} - \bar{g}_{\mu\sigma}\bar{g}_{\nu\rho})$, the linearised spectrum is controlled by four parameters (a, b, c, e) that characterise the linearised field equations [6]:

$$\tfrac{1}{2}\mathcal{E}^L_{\mu\nu} = \big[e - 2\Lambda(a+c) + (2a{+}c)\Box\big]\, G^L_{\mu\nu} + \cdots \tag{9}$$

The physical spectrum is determined by three observables [6]:

$$\kappa_{\rm eff}^{-1} = 4e - 8\Lambda\, a, \tag{10}$$

$$m_g^2 = \frac{-e + 2\Lambda\, a}{2a + c}, \tag{11}$$

$$m_s^2 = \frac{2e - 4\Lambda(a + 4b + c)}{2a + 4c + 12b}. \tag{12}$$

A. Quadratic sector contributions

The Einstein–Hilbert and four-derivative terms contribute

$$\begin{aligned} &\kappa R\text{:} && a{=}0,\ b{=}0,\ c{=}0,\ e{=}\kappa, \\ &\alpha_1 R^2\text{:} && a{=}0,\ b{=}2\alpha_1,\ c{=}0,\ e{=}\tfrac{8\alpha_1\Lambda}{3}, \\ &\alpha_2 R_{\mu\nu}^2\text{:} && a{=}0,\ b{=}0,\ c{=}2\alpha_2,\ e{=}\tfrac{2\alpha_2\Lambda}{3}. \end{aligned} \tag{13}$$

B. Cubic curvature contributions

The cubic invariants $\gamma_i\mathcal{I}_i$ contribute at order Λ to the Bueno–Cano parameters. Writing $a_i = \gamma_i\,\bar{a}_i\,\Lambda$, $b_i = \gamma_i\,\bar{b}_i\,\Lambda$, $c_i = \gamma_i\,\bar{c}_i\,\Lambda$, $e_i = \gamma_i\,\bar{e}_i\,\Lambda^2$, the dimensionless coefficients $(\bar{a}, \bar{b}, \bar{c}, \bar{e})$ are computed numerically via Eqs. (13)–(14) of Ref. [6] and verified analytically for $\mathcal{I}_1$–$\mathcal{I}_4$. The results are collected in Table II.

C. Special limits

The dS spectrum satisfies several non-trivial consistency checks:

TABLE I. The six independent cubic curvature invariants in 4D.

$\mathcal{I}_i$	Definition
$\mathcal{I}_1$	R^3
$\mathcal{I}_2$	$R\,R_{\mu\nu}R^{\mu\nu}$
$\mathcal{I}_3$	$R_\mu{}^\nu R_\nu{}^\rho R_\rho{}^\mu$
$\mathcal{I}_4$	$R\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$
$\mathcal{I}_5$	$R^{\mu\nu}R_{\mu\alpha\beta\gamma}R_\nu{}^{\alpha\beta\gamma}$
$\mathcal{I}_6$	$R_{\mu\nu}{}^{\rho\sigma}R_{\rho\sigma}{}^{\alpha\beta}R_{\alpha\beta}{}^{\mu\nu}$

TABLE II. Bueno–Cano parameters for the six cubic invariants in $D = 4$. The contribution of $\gamma_i\mathcal{I}_i$ to (a, b, c) is proportional to $\gamma_i\Lambda$, and to e proportional to $\gamma_i\Lambda^2$. Computed via finite differences on the parametric Riemann tensor $\bar{R}_{\mu\nu\rho\sigma}(\Lambda, \alpha)$ [Ref. [6], Eq. (11)], verified analytically for $\mathcal{I}_1$–$\mathcal{I}_4$.

	$\bar{a}_i$	$\bar{b}_i$	$\bar{c}_i$	$\bar{e}_i$
$\mathcal{I}_1$	0	6	0	24
$\mathcal{I}_2$	0	1	2	6
$\mathcal{I}_3$	0	0	$\frac{3}{2}$	$\frac{3}{2}$
$\mathcal{I}_4$	4	$\frac{2}{3}$	0	4
$\mathcal{I}_5$	1	0	$\frac{2}{3}$	1
$\mathcal{I}_6$	2	0	0	$\frac{2}{3}$

1. **Flat limit** ($\Lambda \to 0$): all cubic contributions vanish, $m_g^2 \to -\kappa/(2\alpha_2)$ (Stelle), and $\kappa_{\rm eff}^{-1} \to 4\kappa$.
2. **GR limit** ($\alpha_i = \beta_i = \gamma_i = 0$): $a = b = c = 0$, $e = \kappa$, $\kappa_{\rm eff}^{-1} = 4\kappa$, no massive modes.
3. **Einsteinian Cubic Gravity** (ECG) [6]: The unique cubic theory with $2a + c = 0$ and $|m_g^2| = |m_s^2| = \infty$, sharing Einstein gravity's spectrum on any maximally symmetric background. The ECG cubic invariant $\mathcal{P}$ [Ref. [6], Eq. (22)] is a specific linear combination of $\mathcal{I}_1, \ldots, \mathcal{I}_6$.
4. **Conformal gravity** ($\alpha_1 = -\alpha_2/3$, $\beta_i = \gamma_i = 0$): the spin-0 mass diverges, leaving only the massless graviton and a massive spin-2 ghost.

IV. SUMMARY

The complete particle spectrum of six-derivative gravity in 4D is:

Flat space [Eq. (2)]: the propagator is determined by two quadratic form factors $f_2(k^2)$ and $f_0(k^2)$, depending on five parameters $(\kappa, \alpha_1, \alpha_2, \beta_1, \beta_2)$. The cubic couplings γ_i do not contribute.

de Sitter [Eqs. (10)–(12)]: the spectrum is parameterised by $(\kappa_{\rm eff}, m_g^2, m_s^2)$, which depend on all 14

couplings and Λ through the Bueno–Cano parameters (a, b, c, e). The cubic corrections enter at order Λ with the coefficients given in Table II.

The computation was performed entirely within TensorGR.jl and verified against local copies of all cited references. The code, tests (4 569 passing), and reference papers are available in the TensorGR.jl repository.

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